

Nathan Jacobson

COMPUTER SCIENCE STUDENT · DEVELOPER

✉ njacobson604@gmail.com 🏠 <https://nathanjacobson.ca> 🗺 grqphical 🌐 Nathan Jacobson

Experience

Tae Kwon Do Instructor

Port Moody, BC

TRANSCENDING MARTIAL ARTS

September 2025 - Present

- Instructed children aged 4-14 martial arts techniques such as strikes, blocks, patterns, and weapons training
- Lead a group of assistant instructors to aid with instruction and managing the class

Information Officer

Port Moody, BC

ELECTIONS CANADA

April 2025

- Confirmed voter identification
- Directed voter traffic and assisted voters with physical disabilities
- Recorded vote counts after polls had closed

Projects

intellidoc

🗺 grqphical/intellidoc

PYTHON, FASTAPI, SQLITE, LANCEDB

November 2025 - Present

- Currently developing a semantic search web application using Google's EmbeddingGemma model with both vector and keyword retrieval strategies, employing recursive chunking and metadata filtering to maximize accuracy.

interchange

🗺 grqphical/interchange

Go

August 2025-Present

- Created a reverse proxy in Golang that supports proxying, static file hosting, HTTPS, and IP blacklisting/whitelisting
- Currently working on adding rate limiting and load balancing

Skills

Programming Languages

C, Python, Go, HTML, CSS, JS, SQL, Lua

Libraries/Frameworks

pandas, FastAPI, numpy, OpenGL, React, Godot

Other Skills

Vim, Jupyter Notebooks, MS Office, Linux/WSL, Bash

Education

Simon Fraser University

Burnaby, BC

BSc IN COMPUTING SCIENCE

September 2025 - Present

Heritage Woods Secondary

Port Moody, BC

BC DOGWOOD DIPLOMA (HIGH SCHOOL DIPLOMA)

September 2021 - June 2025

- Awarded Top Producing Grade 12 Student
- Graduated with Honour Roll with Distinction

Extra-Curriculars

Tae Kwon Do

SPORT

September 2017 - Present

- Achieved a first degree black belt

Model United Nations

CLUB

September 2023 - June 2025

- Participated in multiple Model UN Conferences (CAHSMUN 2024, VMUN 2025, CAHSMUN 2025)
- Became co-president of my school's Model UN club